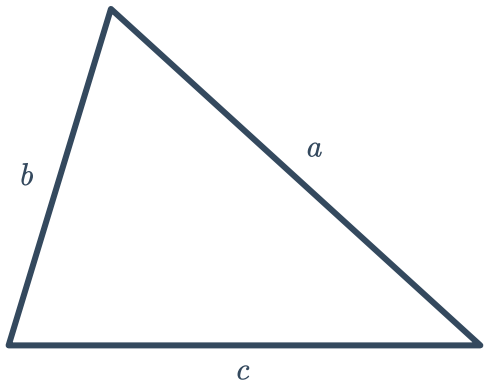
Sine rule

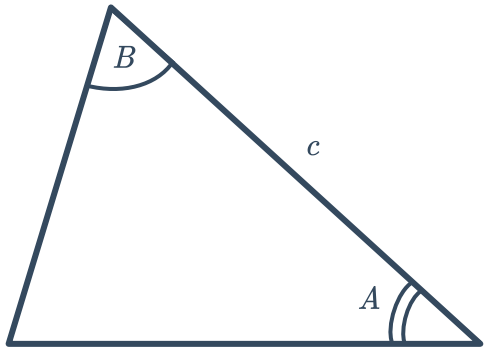
1 An oblique $\triangle ABC$ consists of angles A, B and C which appear opposite sides a, b and c respectively. State whether the following equations are true or false for this triangle:

a $\dfrac{\sin B}{\sin C} = \dfrac{b}{c}$ **b** $\dfrac{a}{\sin A} = \dfrac{c}{\sin C}$ **c** $\dfrac{a}{\sin A} = \dfrac{\sin C}{c}$ **d** $\dfrac{\sin A}{a} = \dfrac{\sin B}{c}$

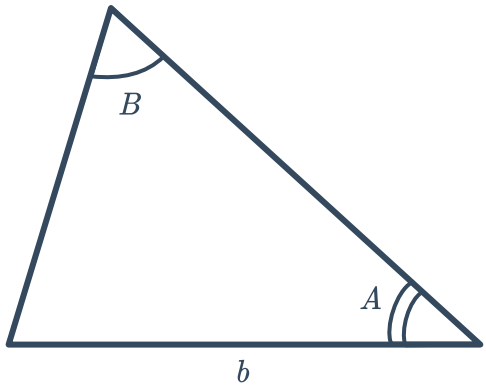
2 Consider a triangle where all three sides are known, but no angles are known. Is there enough information to find all the angles in the triangle using only the sine rule? Explain your answer.



3 Consider a triangle where two of the angles and the side included between them are known. Is there enough information to solve for the remaining sides and angle using the sine rule? Explain your answer.

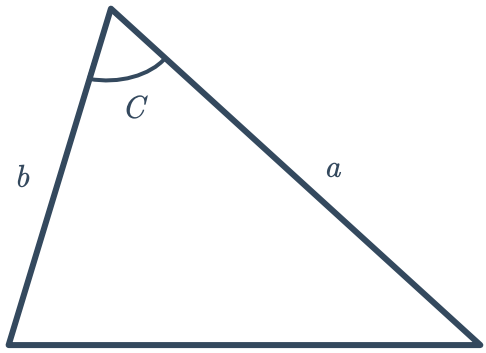


4 Consider a triangle where two of the angles and a side not included between them are known. Is there enough information to solve for the remaining sides and angle using the sine rule? Explain your answer.



© ClipSAT: By Mr. Mohamed Abdallah +966597688647

5 Consider a triangle where two of the sides and an angle included between them are known. Is there enough information to solve for the remaining side and angles using the sine rule? Explain your answer.

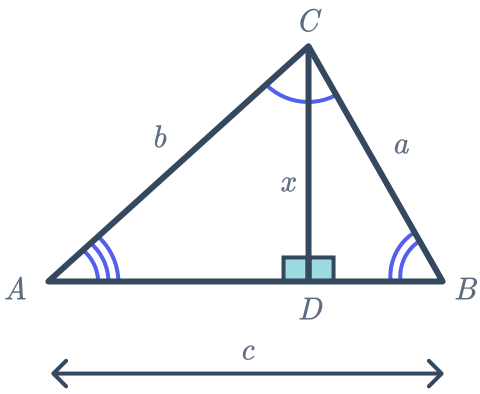


6 Consider the following diagram:

a Find an expression for $\sin A$ in $\triangle ACD$.

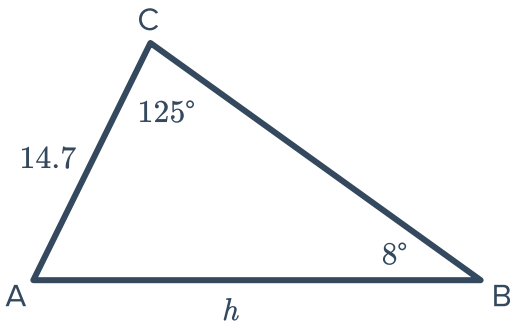
b Find an expression for $\sin B$ in $\triangle BDC$. Then make x the subject of the equation.

c Substitute your expression for x into the equation from part (a), and rearrange the equation to form the sine rule.



Unknown sides

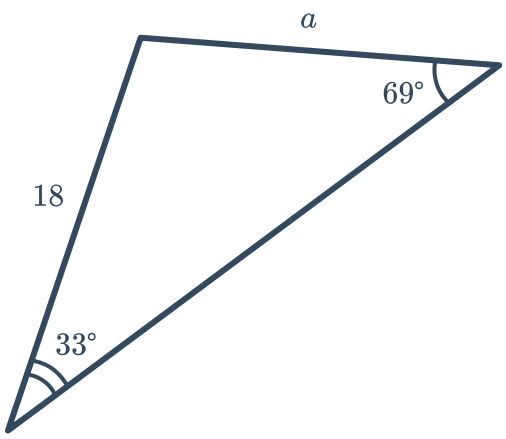
7 Consider the given triangle:
Find the value of h, correct to two decimal places.



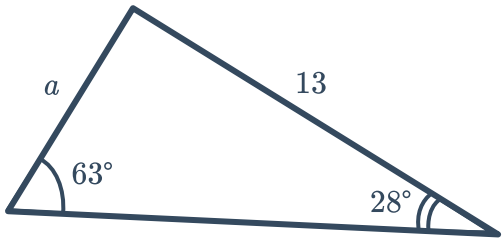
© ClipSAT: By Mr. Mohamed Abdallah +966597688647

8 Find the missing side length of the following triangles using the sine rule, rounding your answers to one decimal place:

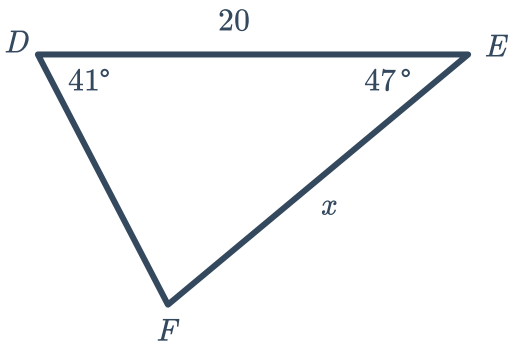
a



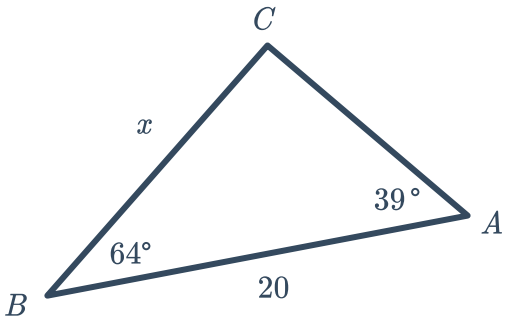
b



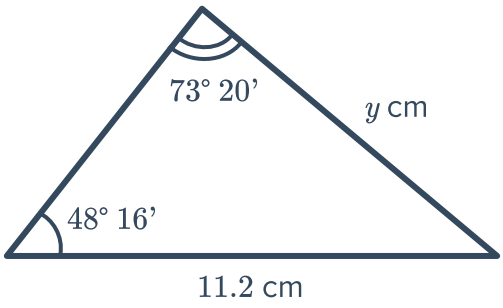
c



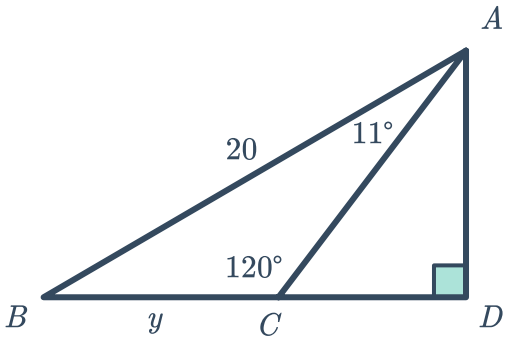
d



e

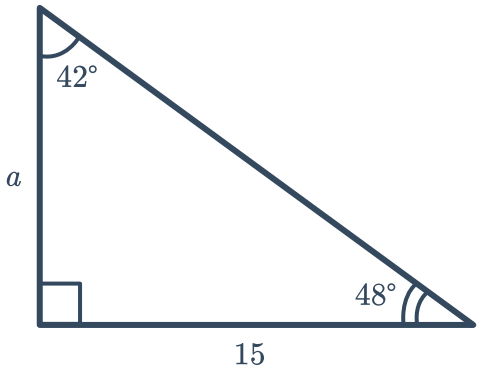


f



© ClipSAT: By Mr. Mohamed Abdallah +966597688647

9 Consider the following triangle:



a Find the length a using the sine rule. Round your answer to two decimal places.

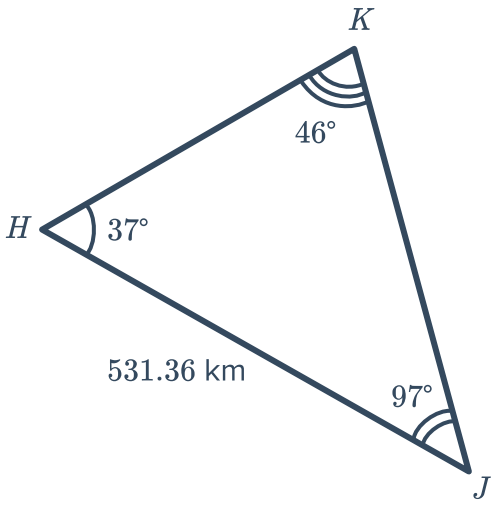
b Use the tangent ratio to confirm the value of a.

10 Triangle ABC has $\angle ABC = 50^\circ 51'$, $\angle ACB = 20^\circ 41'$ and $AC = 15.9$ mm. Find the length of the following, to one decimal place:

a The shortest side.

b The longest side.

11 Consider the following triangle:



a Find the length of side HK, to two decimal places.

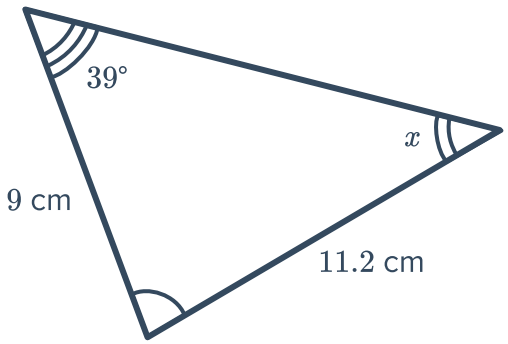
b Find the length of side KJ, to two decimal places.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

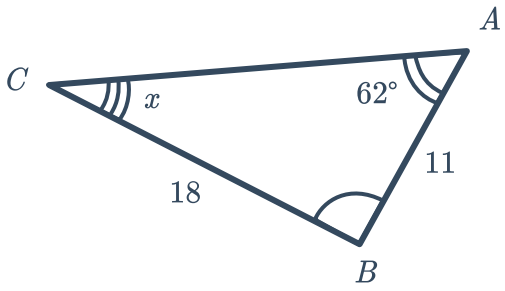
Unknown angles

12 Find the value of the missing angle x, rounded to one decimal place:

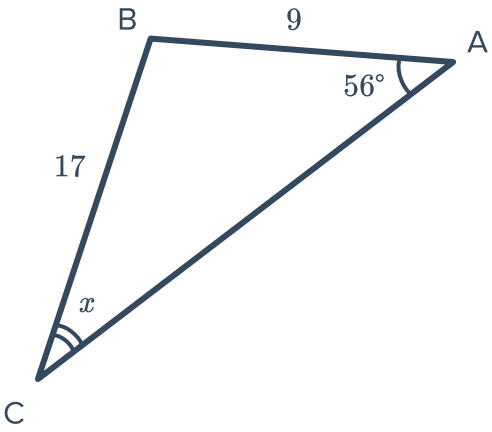
a



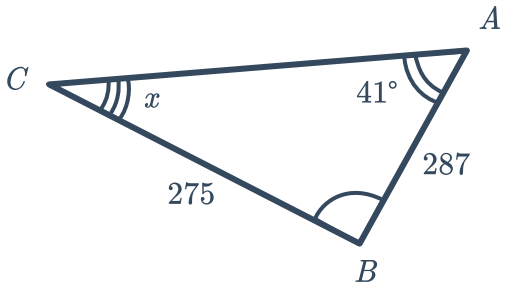
b



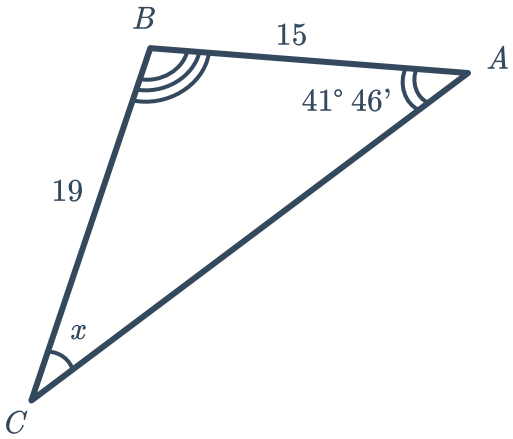
c



d



13 Find the value of x, correct to the nearest minute.



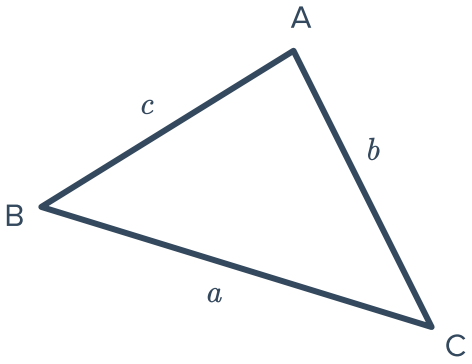
© ClipSAT: By Mr. Mohamed Abdallah +966597688647

14 Consider the triangle with two interior angles $C = 72.53^\circ$ and $B = 31.69^\circ$, and one side length $a = 5.816$ m:

a Find the unknown interior angle A.

b Find b, correct to three decimal places.

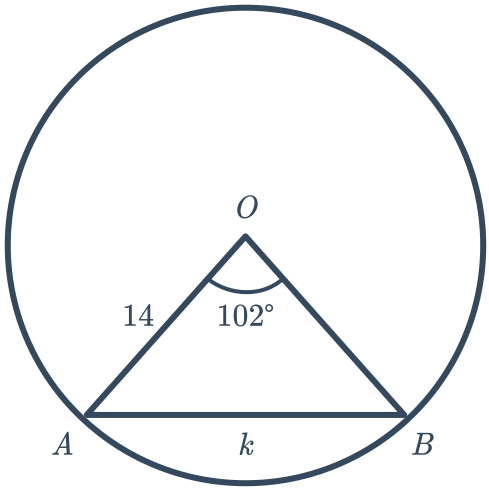
c Find c, correct to three decimal places.



15 Consider the following diagram:

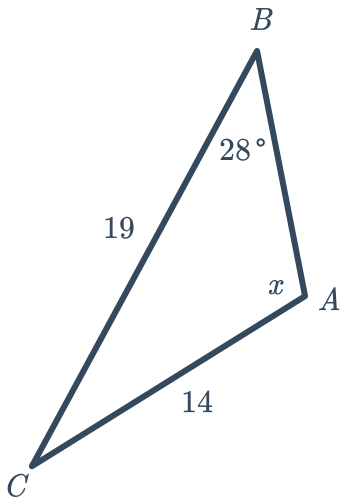
a Find $\angle OBA$.

b Find the length of k. Round your answer to two decimal places.

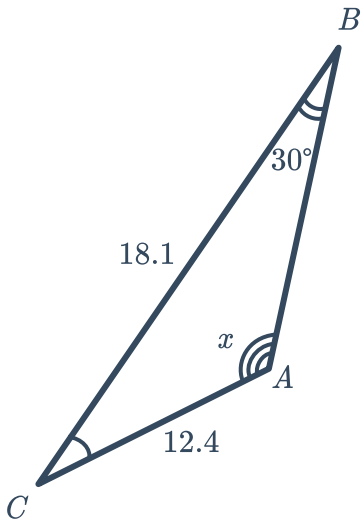


16 Find the size of obtuse angle x for the following triangles, correct to two decimal places:

a

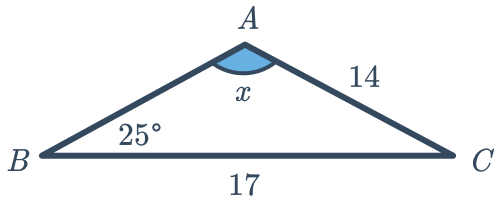


b

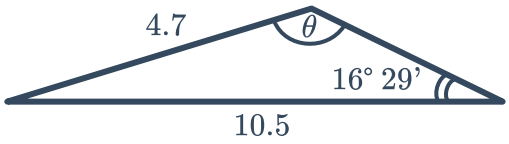


© ClipSAT: By Mr. Mohamed Abdallah +966597688647

c



17 Find the size of the obtuse angle θ, rounded to the nearest minute:



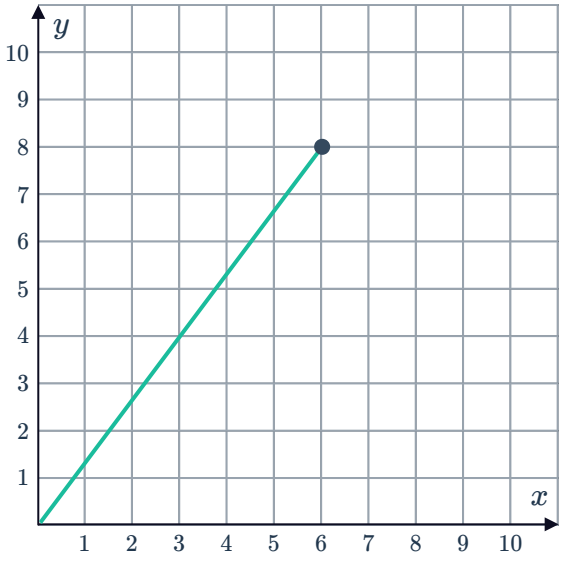
Ambiguous case

18 A line joining the origin and the point $(6, 8)$ has been graphed on the number plane. To form a triangle with the x-axis, a second line is drawn from the point $(6, 8)$ to the positive side of the x-axis.

a In what interval can the length of the second line be such that there are two possible triangles that can be formed with that length?

b What can the length of the second line be such that there is exactly one triangle that can be formed with that length?

c For what lengths of the second line will no triangle be formed?



© ClipSAT: By Mr. Mohamed Abdallah +966597688647

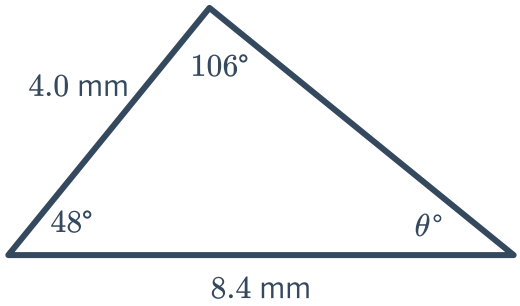
19 Rochelle needs to determine whether a triangle with the dimensions shown below is possible or not:

a Find the value of θ.

b Find the value of $\frac{8.4}{\sin 106^\circ}$. Round your answer to four decimal places.

c Find the value of $\frac{4.0}{\sin 26^\circ}$. Round your answer correct to four decimal places.

d Hence, is it possible to construct this triangle? Explain your answer.



20 $\triangle ABC$ consists of angles A, B and C which appear opposite sides a, b and c respectively. Given the following, state whether solving $\triangle ABC$ results in the ambiguous case:

a If a, b and c are known.

b If A, B and a are known.

c If A, a and c are known.

d If a, B and c are known.

21 Determine the number of possible triangles given the following:

a $a = 50$, $b = 58$ and $A = 60^\circ$

b $a = 23$, $b = 21$ and $A = 35^\circ$

c $B = 30^\circ$, $b = 4$ and $c = 8$

d $a = 39$, $b = 32$ and $B = 50^\circ$

22 Determine whether the following sets of data determine a unique triangle:

a $B = 40^\circ, b = 2, c = 5$

b $a = 6, b = 3, c = 27$

c $a = 3, b = 4, c = 5$

d $a = 80^\circ, b = 20^\circ, c = 80^\circ$

e $a = 5, b = 6, C = 80^\circ$

f $A = 50^\circ, B = 30^\circ, c = 8$

g $a = 20^\circ, b = 40^\circ, c = 120^\circ$

h $a = 5, b = 12, c = 13$

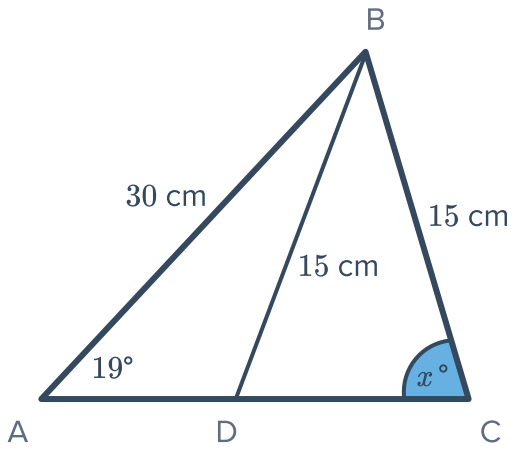
i $\angle CAB = 42^\circ, a = 7, b = 2$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

23 Consider $\triangle ABC$ below:

a Find x, given that x is acute. Round your answer to the nearest degree.

b Find $\angle ADB$ to the nearest degree, given that $\angle ADB > x$.



24 For each of the given measurements of $\triangle ABC$:

i Determine whether such a triangle exists.

ii If so, state whether the triangle could be acute and/or obtuse.

a $\angle CAB = 36^\circ$, $a = 7$ and $b = 10$

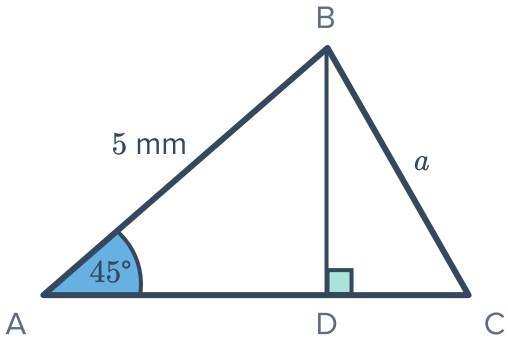
b $\angle CAB = 35^\circ$, $a = 5$ and $b = 11$

25 $\triangle ABC$ is such that $\angle CAB = 32^\circ$, $a = 5$ and $b = 9$.
Let the unknown angle opposite the length 9 cm be x.

a Consider the acute case, and find the size of angle x, to two decimal places.

b Consider the obtuse case, and find the size of the obtuse angle x, to two decimal places.

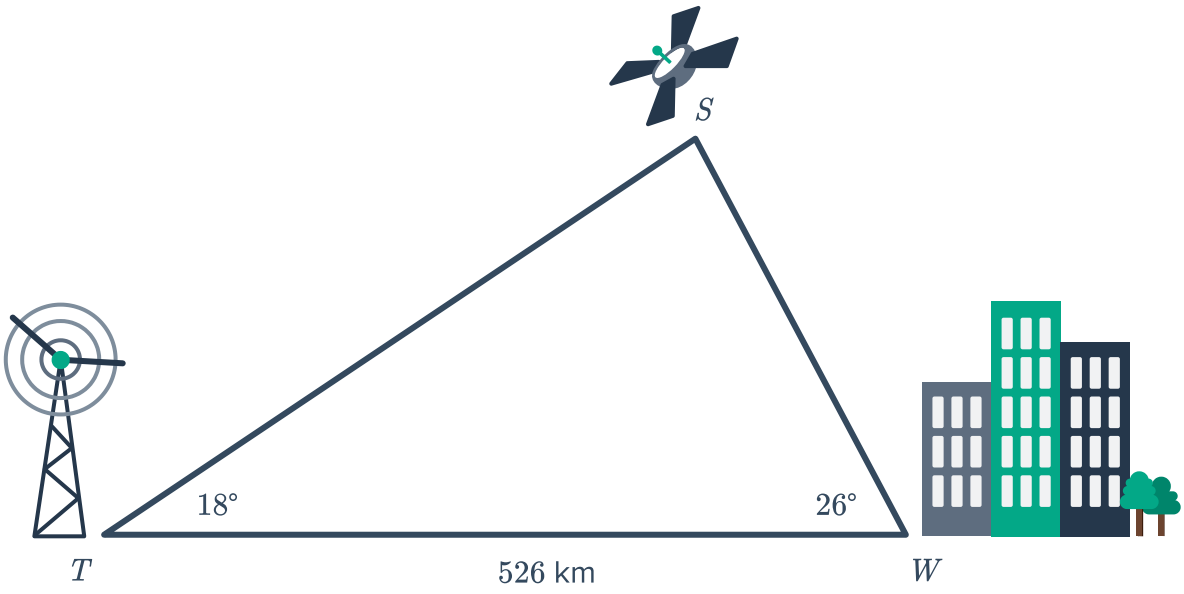
26 In $\triangle ABC$, $A = 45^\circ$ and $c = 5$ mm.
What is the range of lengths, rounded to the nearest tenth where appropriate, for BC that lead to the ambiguous case where we don't know if the triangle formed is acute or obtuse?



© ClipSAT: By Mr. Mohamed Abdallah +966597688647

Applications

27 A radio signal is sent from a transmitter at tower T, via a satellite S, to a town W, as shown in the diagram. The town is $526\ \text{km}$ from the transmitter tower. The signal is sent out from the transmitter tower at an angle of 18°, and the town receives the signal at an angle of 26°.

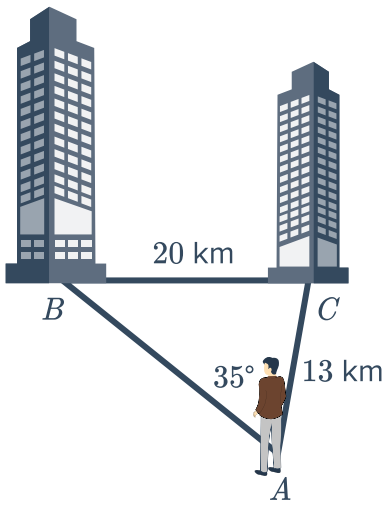


a Find the size of $\angle WST$.

b Find the distance, SW, that the signal travels from the satellite to the town. Round your answer to the nearest kilometre.

c If the satellite is $h\ \text{km}$ above the ground, find h. Round your answer to two decimal places.

28 Dave is standing on a hill and can see two buildings in the distance. Suppose the buildings are $20\ \text{km}$ apart. Dave is $13\ \text{km}$ from one building and the angle between the two lines of sight to the buildings is 35°.



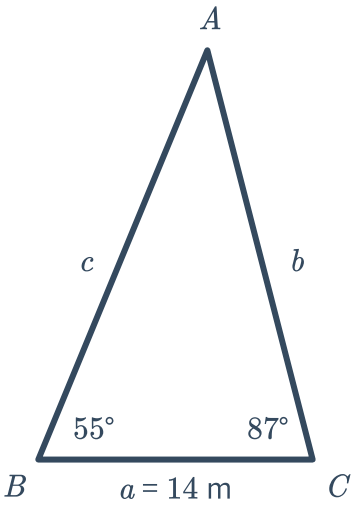
a Find $\angle ABC$. Round your answer to two decimal places.

b Find $\angle BCA$. Round your answer to two decimal places.

c Find the distance from Dave to the building at B. Round your answer to one decimal place.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

out the perimeter of a triangular course that players need to run around. The diagram shows some measurements taken of the course, where side length $a = 14$ m:



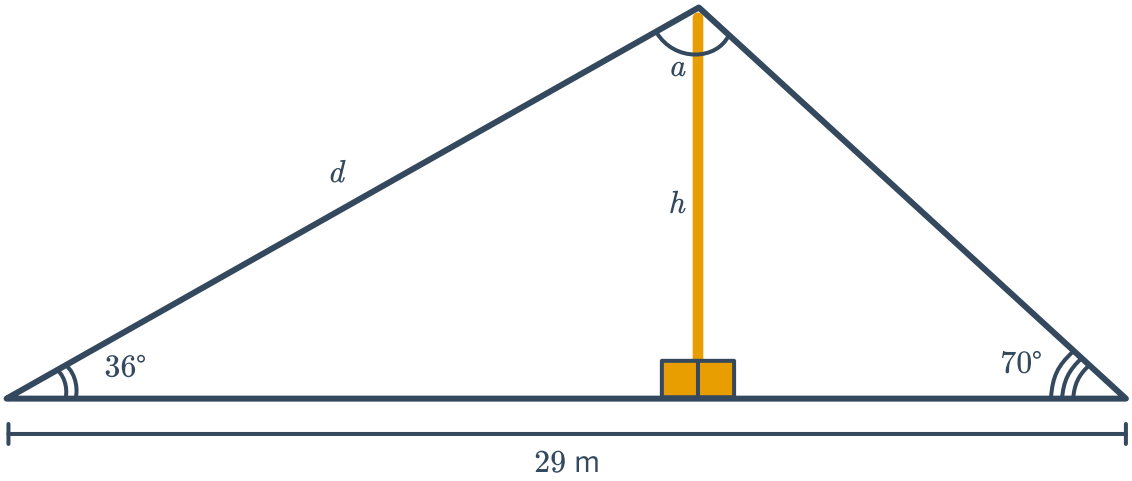
a Find $\angle A$.

b Find the length of side c. Round your answer to two decimal places.

c Find the length of side b. Round your answer to two decimal places.

d Each player must sprint one lap around the triangle and then jog one lap around the triangle. This process is to be done 3 times by each player.
If Tara can run $280\ \text{m/min}$, and can jog at half the speed she runs, how long will this exercise take her? Round your answer to one decimal place.

30 Two wires help support a tall pole. One wire forms an angle of 36° with the ground and the other wire forms an angle of 70° with the ground. The wires are 29 m apart.



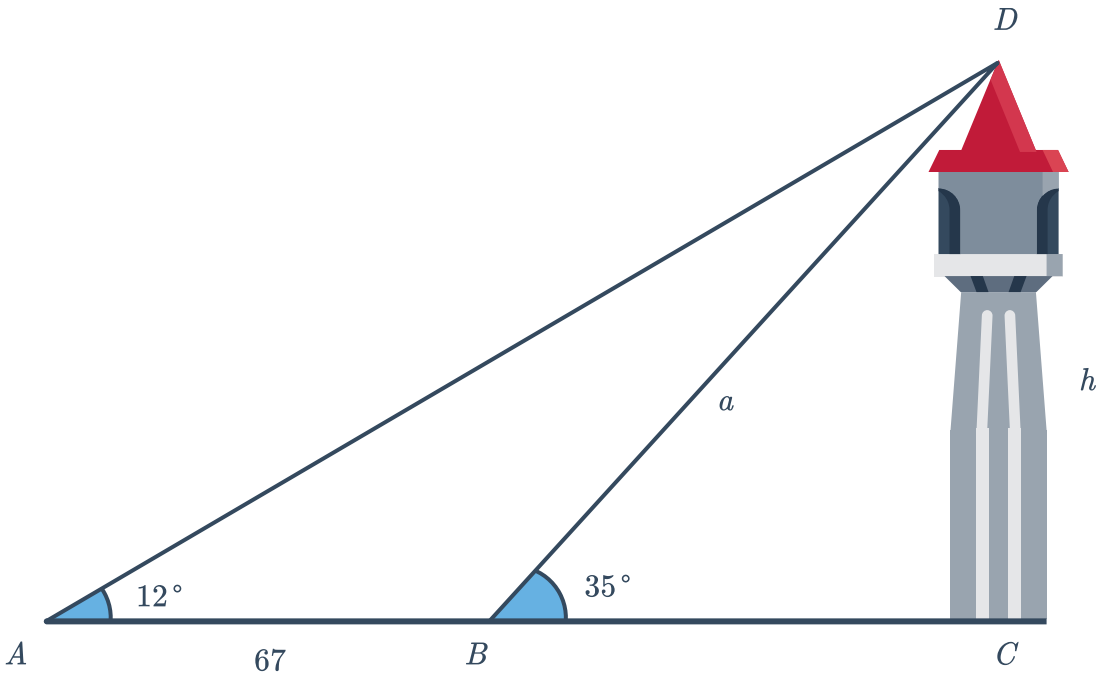
a Find a, the angle made between the two wires at the top of the pole.

b Find d, the length of longest wire in metres. Round your answer to two decimal places.

c Calculate h, the height of the pole in metres. Round your answer to two decimal places.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

31 Mae observes a tower at an angle of elevation of 12°. The tower is perpendicular to the ground. Walking 67 m towards the tower, she finds that the angle of elevation increases to 35°.



a Calculate the angle $\angle ADB$.

b Find the length of the side a. Round your answer to two decimal places.

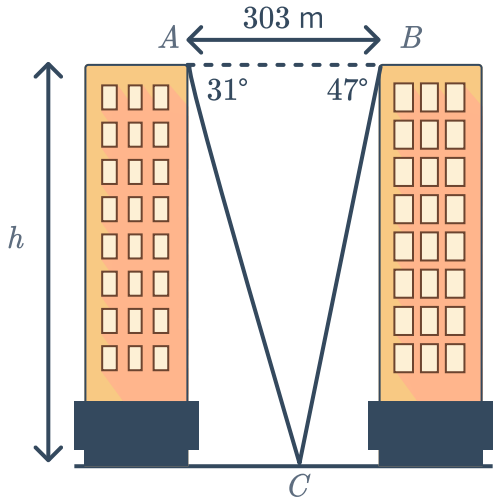
c Hence, find the height h, of the tower. Round your answer to one decimal place.

32 To calculate the height of each block of flats, a surveyor measures the angles of depression from A and B to C. From A, the angle of depression is 31° and from B the angle of depression is 47°.

a Find the size of $\angle ACB$.

b If the distance between A and C is b m, find the value of b. Round your answer to two decimal places.

c If the buildings are h m tall, find the value of h. Round your answer to the nearest metre.



© ClipSAT: By Mr. Mohamed Abdallah +966597688647

33 Competitors taking part in a fundraising event must make their way around a triangular course set up in open water. They must swim from buoy A to buoy B, stand-up paddle from buoy B to buoy C, and then kayak from buoy C back to buoy A. The buoys are set up such that $\angle CAB = 61^\circ 17'$ and $\angle ABC = 73^\circ 12'$. The swimming leg is $250\ \text{m}$ long.

a Find BC, the length of the stand-up paddling leg of the course, correct to one decimal place.

b Find AC, the length of the kayaking leg of the course, correct to one decimal place.

c If the maximum time possible to finish the course is 22 minutes, find the slowest possible average speed of a competitor throughout the course. Round your answer to the nearest metre per minute.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647